

Architecture Decision Records

Short, dated records of decisions that shape Git-Vista's architecture — the kind of thing that is expensive to reverse and easy to forget the *why* of. Each ADR captures the context, the decision, the alternatives weighed, and the consequences, so a later reader (or a later us) can see not just what was chosen but what it was chosen over.

One file per decision, numbered in order: `NNNN-short-slug.md`. ADRs are append-only history — supersede an old one with a new one rather than rewriting it, and note the link in both.

ADR	Title	Status
0001	Repository identity and the repository-generation algorithm	Accepted
0002	A versioned API contract: protocol negotiation, structured errors, and a transport crate	Accepted
0003	A server-owned, allowlisted repository catalog addressed by opaque id	Accepted
0004	Protecting loopback sessions and mutating requests	Accepted

ADR	Title	Status
0005	LAN view profile: a read-only second listener	Accepted — implemented
0006	Visualize / Active is chosen per open, every time	Accepted — implementation pending
0007	Mode rides the current-repo selection (<code>POST /api/select</code>)	Accepted — implementation pending
0008	Persistent, multiple clones under the XDG data dir	Accepted
0009	Local repos discovered from one configured root, direct children only	Accepted — implementation pending
0010	Visualizer = existing read-only views plus forge deep links	Accepted — implementation pending
0011	Gesture slop is pointer-type-aware (touch 12px, mouse/pen 4px)	Accepted
0012	The app shell never scrolls; all scrolling is internal	Accepted
0013	Protocol v2 for the clone response; <code>set_current</code> returns its own handle	Accepted

ADR	Title	Status
0014	LAN-view sessions present read-only affordances only	Accepted
0015	A closed Git-operation vocabulary and the reviewable Plan schema	Accepted
0016	Every write action executes through one shared planner	Accepted
0017	No arbitrary argv from the browser, held closed by a tripwire	Accepted
0018	Stale, tampered or expired plans never execute	Accepted
0019	One mutation at a time per shared repository	Accepted
0020	Idempotent operation lifecycles and reconnectable progress	Accepted
0021	Durable operation journal and recovery references	Accepted
0022	Paged history, signed cursors, and bounded reads	Accepted
0023	Rehearsal workspaces promote results, atomically	Proposed

ADR	Title	Status
0024	Frontend overlay state moves into a Dock -keyed OverlayStack	Accepted
0025	Hook policy: a declared, disclosed value, not yet enforced	Accepted
0026	ShellMode : Rust owns the layout mode, CSS keys off one class	Accepted
0027	Secrets inside a granted \$HOME are withheld by enumerate-and-skip, not a deny rule	Accepted
0028	Accept that the network tier constrains ports, not hosts (Option A)	Accepted
0029	INV-13: hard-fail when the Strict tier is selected but unavailable	Accepted — implementation pending
0030	The git-process sandbox: a pure argv boundary, tiers by declared intent, and tests that prove their own premise	Accepted — core mechanism and INV-15 disclosure landed and tested

ADR	Title	Status
0031	Every ADR records its alternatives and why they lost	Accepted
0032	No service worker: offline is a failure to surface, not to mask	Accepted
0033	SSH remotes under the sandbox: a narrow, explicit carve-out through <code>secret_excludes</code>	Accepted — implemented and tested
0034	File-at-commit reads go through one long-lived <code>cat-file --batch</code> process	Accepted — implemented and tested
0035	The inspector bottom sheet is wired to <code>ShellMode</code> , following the finger during drag	Accepted — implemented and tested
0036	The Network-tier exec harness: forced askpass hardening, byte-level redaction, and what stays open	Accepted — implemented and tested
0037	Observe state, never parse git's prose: destructive operations report what the worktree proves	Accepted — implemented and tested

ADR	Title	Status
0038	Worktree-destructive operations: typed per-path impact, a preview that names every file, and one required control that cannot exist	Accepted — implemented and tested
0039	The typed remote-operation vocabulary: <code>FetchRemote</code> , <code>PullBranch</code> , and a lease-guarded <code>PushBranch</code>	Accepted — typed contract implemented and tested; execution wired for all three by ADR 0043 (#229), ADR 0044 (#230) and ADR 0045 (#231)
0040	Amend execution: its own route, an executor-level CAS, an advisory published-history flag, and typed failure kinds	Accepted — implemented and tested
0041	The typed tag vocabulary: four variants, and an undo that restores the exact tag object	Accepted — typed contract implemented and tested; local execution wired by 0048 , remote pair still refused (M2.21 slices of #74)
0042	The planner's build / submit seam: two stages, one set of stage functions	Accepted — implemented and tested; not routed until #248/#249

ADR	Title	Status
0043	Fetch execution: progress on the lifecycle that already exists, a cancel that kills the child, and an outcome read from refs rather than prose	Accepted — implemented and tested; its claim about <i>which host</i> a fetch may reach is corrected by ADR 0047
0044	Pull execution: one fetch in the server, a strategy the wire must state, and a conflict that is an outcome rather than an error	Accepted — implemented and tested
0045	Push execution: a force that cannot be built, a lease checked by two parties, and a cancel that refuses to reassure	Accepted — implemented and tested
0046	The MCP plan-tool surface: 23 build-only tools, one endpoint, and a variant that cannot be exposed	Accepted — implemented and tested
0047	Which host a fetch may contact: a name-shaped newtype <i>and</i> a precondition that refuses instead of being skipped	Accepted — implemented and tested

ADR	Title	Status
0048	Local tag execution: the annotation that cannot be empty, and the pin that outlives the tag	Accepted — implemented and tested; signing and the two remote-reaching tag operations still refuse (M2.21e/f of #74)
0049	V1 scope freeze: eighteen never-started issues closed as won't-do, M6/M7 retired, M8 deleted	Accepted — tracker changes only; no code or protocol touched
0050	Learning an operation's id before it finishes: an additive GET <code>/api/operations/by-key/{key}</code>	Accepted — implemented and tested
0051	Intent admission belongs <i>after</i> every await, not once before them	Accepted — implemented, confirmed on a device
0052	An explicit repo list, because "these four" is not "this folder"	Accepted — implemented and tested
0053	"Cancellable" is satisfied by the framework and an id echo, not by a tracker of our own	Accepted — implemented

ADR	Title	Status
0054	The Linux desktop browser is the verification target; iPad deferred to a VNC display	Accepted
0055	Working-tree status readings carry a server-stamped <code>scanned_at</code> , additive to the v1 wire contract	Accepted — implemented
0056	<code>gv --repo</code> registers a catalog entry via the boot-time <code>GIT_VISTA_REPOS</code> unit var, not the live <code>/api/clone</code> endpoint	Accepted — implemented
0057	The commit draft moves to <code>localStorage</code> , offered back through an aged restore banner, never auto-filled	Accepted — implemented
0058	Commit-path git spawns that run hooks are time-bounded, killed, and verified	Accepted — implemented
0059	Plain-commit failures get a typed <code>CommitFailureKind</code> , split finer than amend's on signing	Accepted — implemented

ADR	Title	Status
0060	<code>refuse_if_git_busy</code> verifies liveness via <code>/proc</code> before trusting <code>index.lock</code> , and removes a lock confirmed stale	Accepted — implemented
0061	<code>Plan</code> carries <code>advisories</code> ; a force-with-lease names the default branch, says when it could not tell, and states that the remote cannot be undone	Accepted — implemented
0062	A two-endpoint <code>DiffSpec</code> carries an explicit basis (two-dot vs three-dot); reversal preserves it, and only a <code>Direct</code> reversal is an inverse	Accepted — implemented
0063	One conflict vocabulary for all six operations; each side is <code>Present</code> , <code>Absent</code> or <code>Unreadable</code> , and none may collapse	Accepted — implemented
0064	Resolving a conflict is a <code>GitOperation</code> ; no precondition can express "still conflicted", so the executor re-reads and refuses	Accepted — implemented

ADR	Title	Status
0065	<code>set +e</code> for recording left errexit off inside <code>gate_body</code> , so the gate could not fail for three days; errexit is re-armed and a test drives the real script to prove it	Accepted — implemented
0066	Inspecting a conflict is three reads the LAN listener never sees; the pane mapping is host-tested so "absent" cannot quietly render as empty	Accepted — implemented
0067	One path per resolve, the refusal rendered inline as the answer, and a path validated by the wire format rather than by a handler that could forget	Accepted — implemented

ADR	Title	Status
0068	A conflict's shape is read from kind , never inferred from the deletion flags — which conflate "never had it" with "deleted it"; controls that cannot succeed are replaced by their reason; rename detection recorded as unbuildable	Accepted — implemented
0069	Block/line conflict resolution seeds its editor from the working-tree marker file, not composed panes, and a new <code>conflict-v1:</code> token pins exactly what was served — closing the one blind spot no existing staleness check covers	Accepted — design only, no code

ADR	Title	Status
0070	A ref capture says which kinds it recorded, and stays silent about the rest — HEAD, tags and remote-tracking refs join the per-event snapshot, each new field Option so a pre-#449 line's silence is never read as "there were none"	Accepted — implemented and tested
0071	A badge is a claim about a commit — HEAD is badged only when it resolves, so two readers stop contradicting each other; a dangling HEAD is recorded as Unresolvable rather than drawn as a badge pointing at nothing	Accepted — implemented and tested
0072	HEAD's state is said on the wire, not inferred from an absent branch name — head_branch: null covers a healthy detached HEAD and a broken one alike, and the topbar drew nothing for both	Accepted — implemented and tested

ADR	Title	Status
0073	A sign-in link pasted into a live tab reloads the page rather than re-bootstrapping in place — a fragment edit fires only <code>hashchange</code> , and per-tab session facts are fixed once <code>establish_session</code> resolves	Accepted — implemented and tested
0074	A diagnostic may not fabricate — doctor's clones root is read from the running listener's own <code>/proc/<pid>/environ</code> , says unknown when it cannot be read, and says whether it exists; a fabricated line inherits the credibility of the measured one beside it	Accepted — implemented and tested

ADR	Title	Status
0075	A WIP-checkpoint run is identified by ancestry, never by proximity — a branch and its diverged twin interleave row for row, so a scan over display neighbours is only ever shown cross-chain pairs; relaxing the lane check to fix that was refused as a quiet lie	Accepted — implemented and tested
0076	One fixture catalogue, in Rust, shelled out to by the browser harness — twenty hand-rolled <code>seeded_repo()</code> builders and three conflicted-repository builders collapse into one crate whose doc comments are the teaching material, so a lesson that drifts from the code fails a test	Accepted — implemented and tested; browser leg unrun

ADR	Title	Status
0077	A pop is two operations, and its report never outruns what was checked — the drop runs only on an applied stash plus a conflict scan that really ran; a refused apply is not proof nothing was applied, so the scan is consulted on both outcomes; exactly one of seven verdicts claims completion	Accepted — implemented and tested; the browser leg has since been run and found three defects the slice shipped with (see the ADR)

ADR	Title	Status
0078	There is one pop, it is composed, and apply asks before it claims — the fully wired but unreachable PopStash is deleted so the "deliberately ABSENT" comment beside it becomes true, and exec_apply_stash re-reads the conflict state on both outcomes, split into a pure verdict and a pure rendering so the combinations real git will not produce are unit-tested rather than recorded as a gap; the HTTP status mirrors git's exit status and never what the scan found, because ApplyOutcome and the durable operation row both read it and neither can read a body	Accepted, with a same-day correction: it was NOT unreachable — the generic /api/plan + /api/execute-plan seam built and ran one, so the deletion closed a live path rather than removing dead code (found by codex)

ADR	Title	Status
0079	The stash endpoints share their DTOs, and the wire carries the selector rather than the position — four shapes move to <code>git-vista-protocol</code> with the newtypes doing the validating, so the handlers have no validation left to delete; the derived <code>index</code> field leaves the listing because a field derivable from another is a second place for one fact to be wrong	Accepted — implemented and tested; browser leg unrun
0080	A journal line may point at its batch's capture, and nothing leaves the server as a pointer — one operation reads the refs once, the batch's last line stores them, and <code>assemble_feed</code> resolves the reference before the feed goes out; bytes/line at 500 refs falls from 29,350 to 225	Accepted — implemented and tested

ADR	Title	Status
0081	An admission that git could not be read is not a ref, and never folds into a count — the "tips unknown" entry was counted as the fifth of four moved refs and erased in the process; the discriminator is the whole all-None shape, and both halves of the conjunction are pinned by their own synthetic fixture	Accepted — implemented and tested

ADR	Title	Status
0082	A documented version floor is exercised, not asserted — CI builds the real git 2.32 named by <code>SUPPORTED_VERSIONS.md</code> and runs the whole status vocabulary through the parser with both binaries; because the expected answer is "no difference", whether the second leg ran is asserted in shell over the test's own report rather than by the test, which may neither skip nor fail when the binary is absent	Accepted — implemented and tested
0083	Test repository selection is task-local while production remains process-global — selection-writing tests enter an enforced scope, and the planner explicitly carries that scope across its detached task boundary	Accepted — implemented and tested

ADR	Title	Status
0084	One error-rewrap mechanism, and it forwards what it cannot buffer — <code>rewrap_error</code> splits the body at its 64 KiB cap instead of collecting it all-or-nothing, so an over-cap refusal is streamed on whole rather than replaced by "Bad Request"; the route-local layer that alone survived that edge also alone labeled its route's 200, so the same mechanism now labels every hand-serialized success too — gated on the body's own <code>size_hint</code> so the progress stream is never polled	Accepted — implemented and tested; browser leg unrun

ADR	Title	Status
0085	A journal line is a versioned envelope, and "I cannot read this capture" is an answer — #485's <code>in_batch</code> is unreadable to a pre-#485 binary and takes the whole event with it, so a rolled-back 100-ref fetch renders as one row; that cost is accepted and written into ADR 0080, while new lines carry a per-line format stamp and <code>RefsAtEvent</code> gains a catch-all, so the <i>next</i> format change costs a line its capture rather than the line and the reader names the version instead of guessing	Accepted — implemented and tested; browser leg unrun

ADR	Title	Status
0087	The CI floor build pins a reviewed commit, not a mutable tag, and checks it after the clone but before make — the cache key gains that commit plus the runner's arch/toolchain identity, so no pre-pin artifact can be served under the trusted key; the reconciliation of codex's reasoned finding against the local review that found nothing is recorded in full	Accepted — implemented; dry-run in the PR, live on the PR's own CI

ADR	Title	Status
0088	A signature status says what the crypto said — EXPKEYSIG , EXPSIG and the unmatched REVKEYSIG stop collapsing into "valid" and "we could not check"; three flat variants keep every consumer's match exhaustive, the classifier resolves by an explicit most-alarming-first table, and the fallthrough keeps a named census so the next unmodelled gpg status is loud rather than invisible; adding a variant to a closed vocabulary fails the whole record at an older client, so the window moves to protocol 8	Accepted — implemented and tested; browser leg unrun

ADR	Title	Status
0089	An undecodable operation row is incompatible, never missing — "cannot decode" and "does not exist" become different answers at every caller, a stranded row is force-failed instead of running forever, its idempotency key can no longer be spent on a fresh execution, and only a payload that would not deserialize may be blamed on another build (five of the six decode-failure arms cannot observe version skew, and the claim was being written permanently to disk)	Accepted — implemented and tested

ADR	Title	Status
0090	A drop that completes a pop proves the tree still holds what the apply restored — the entry not having moved was never evidence the changes were still there, and a <code>reset - -hard</code> between a pop's two halves deleted the stash over a wiped tree while reporting success; the drop now names the apply it completes and the server compares the generation that operation recorded against a live read, inside the coordinator guard, because the plan is built before the guard is taken and anything checked earlier is looking at a repository another writer can still change	Accepted — implemented and tested; protocol v9

ADR	Title	Status
0091	What an operation does to your files is derived, not declared — index/worktree and remote effects become exhaustive accessors on the closed <code>GitOperation</code> vocabulary rather than new <code>Plan</code> fields, so #92 costs no wire change and no version gate; <code>NetworkNeed</code> and its typed classifier move into the protocol crate as pure motion while the <code>argv</code> classifier stays behind as a sandbox guard; where an effect is not a function of the operation the answer names the condition instead of worst-casing it; and the parity test's derived half is anchored on a hand-written table because calling the accessor would assert $f(x) == f(x)$	Accepted — implemented and tested; viewer half not yet built